

AI CHATBOT FOR COLLEGE — CHANGED PAGES ONLY (LATEST REVISION)

Interaction boundaries and broad secondary database cylinder actor

1. Use Case Diagram (Broad Database Cylinder)

✓ Teacher Correction Applied: Broad database cylinder with zero stickman icon.

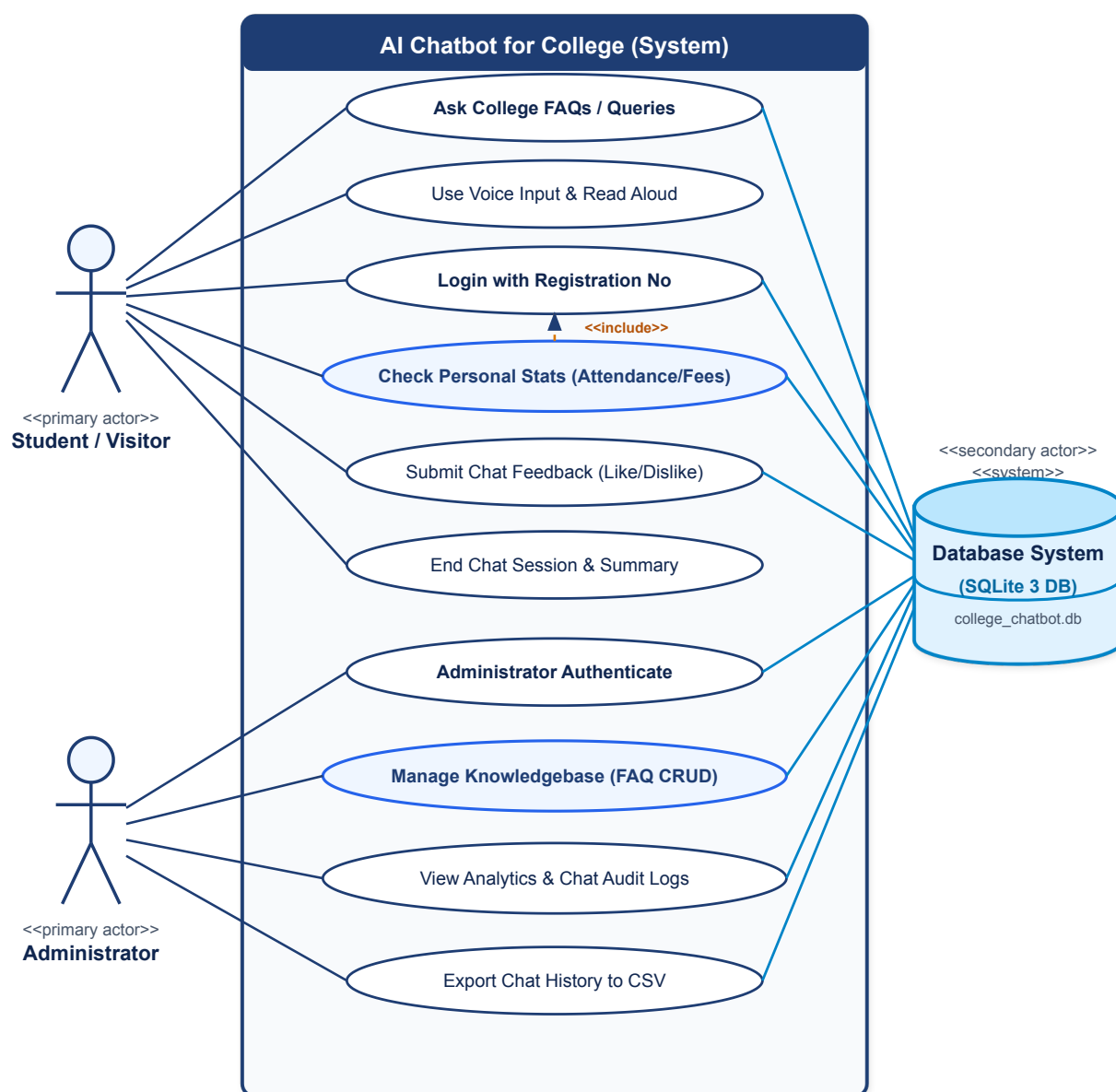


Fig 1: Use Case Diagram — primary actors and broad secondary database cylinder.

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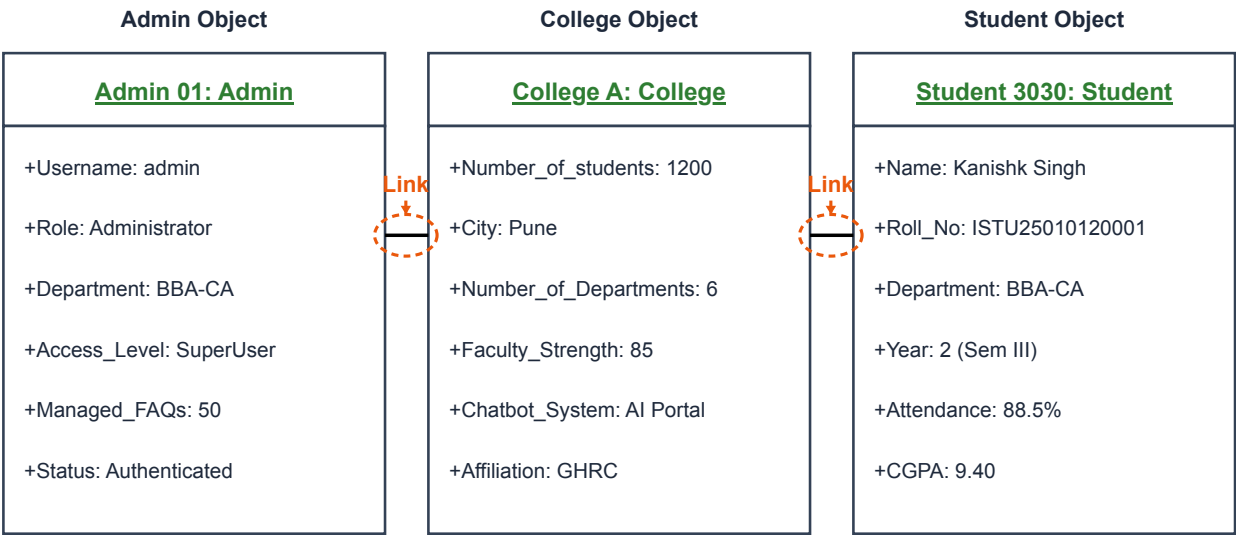
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Runtime execution memory snapshot matching GeeksforGeeks reference format

2. Object Diagram (Admin, College & Student Objects)

✓ Teacher Correction Applied: Admin, College, and Student objects with + slots and link associations.

An object diagram using links and 3 objects



An object of class Admin and an object of class Student are linked to an object of class College.

Detailed Object Diagram Breakdown (3 Concrete Instances):

- Admin Object (Admin 01: Admin): Models administrator authentication, department role, and active session permissions.
- College Object (College A: College): Central institutional node managing student capacity (1200), department quotas, and AI Chatbot.
- Student Object (Student 3030: Student): Concrete student instance Kanishk Singh (ISTU25010120001) with 88.5% attendance and CGPA.
- Link Semantics: Both Admin and Student are connected via formal binary links to the College entity, representing institutional association.

Fig 2: Object Diagram — Admin, College, and Student objects connected via formal links.

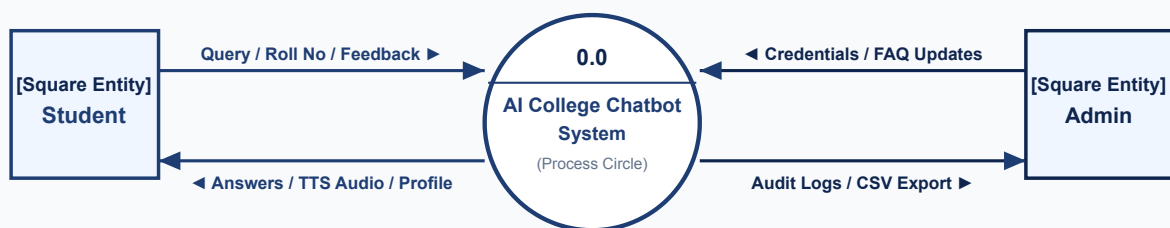
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Information flow with square entities and circle processes

3. Data Flow Diagram — DFD (Square Entities & Circles)

✓ Teacher Correction Applied: Square external entities & circle process symbols with solid flow arrows.

DFD Level 0: Context Diagram



DFD Level 1: Process Decomposition (Circles & Square)

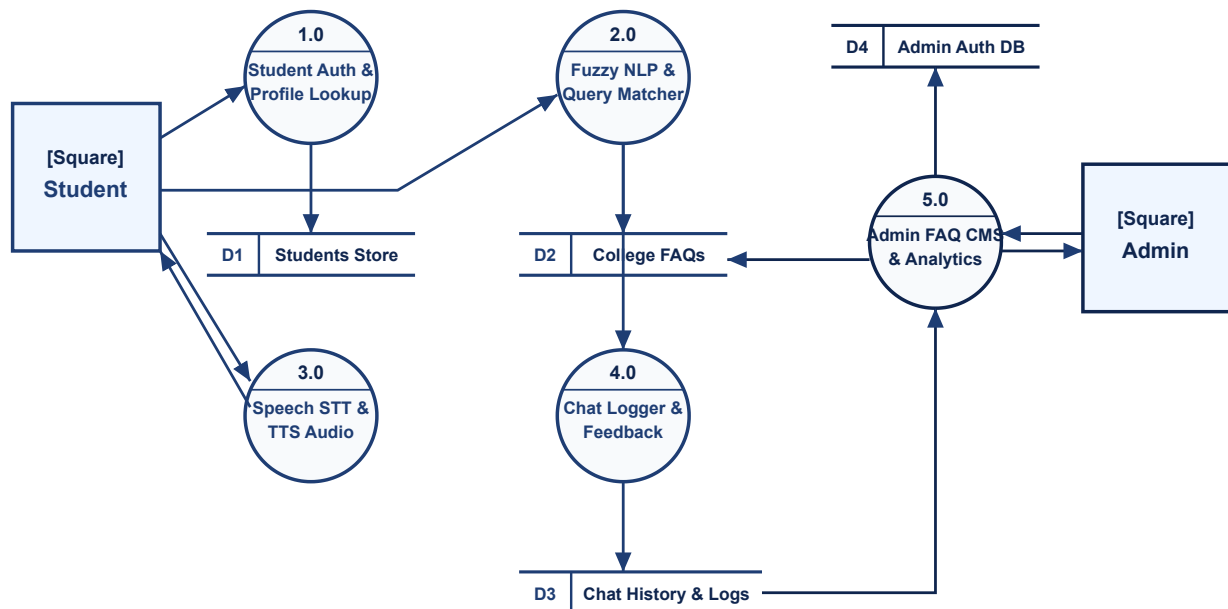


Fig 3: Data Flow Diagram (DFD) — square external entities and circle processes.

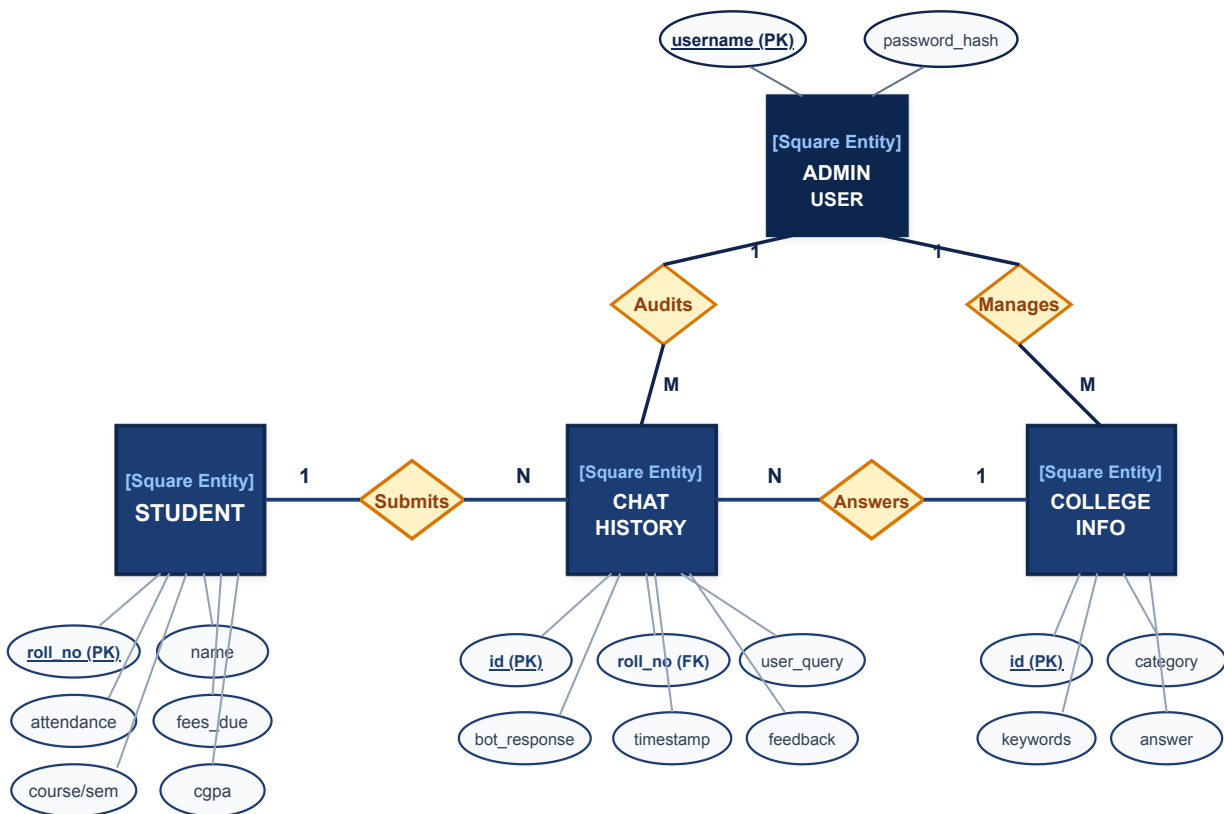
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Relational data schema with square entities and cardinalities

4. Entity-Relationship Diagram — ERD (Square Entities)

✓ Teacher Correction Applied: Square entity symbols for all 4 tables with clean non-crossing flow.



✓ Standard Entity-Relationship Diagram Flow Summary:

- **Exact Square Entity Symbols:** All entities (STUDENT, ADMIN_USER, CHAT_HISTORY, COLLEGE_INFO) rendered as square boxes.
- **Left-to-Right Primary Data Flow:** STUDENT [1] submits [N] CHAT_HISTORY, which answers [N:1] from COLLEGE_INFO.
- **Top-to-Bottom Admin Control Flow:** ADMIN_USER [1] audits [M] CHAT_HISTORY and manages [M] COLLEGE_INFO FAQs.
- **Relational Integrity:** Primary Keys underlined (roll_no, id, username); Foreign Key marked (roll_no FK).

Fig 4: Entity-Relationship Diagram (ERD) — square entities and clean relational flow.

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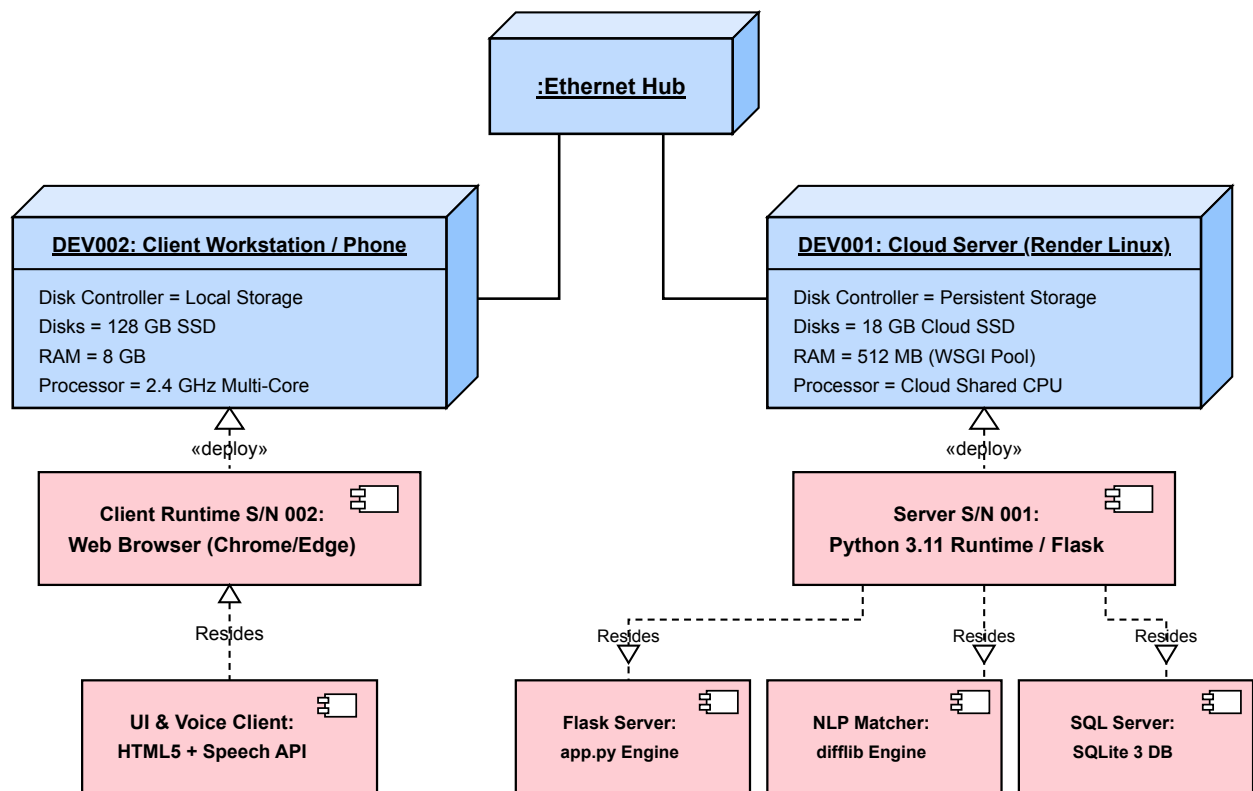
Development Environment Model matching reference architecture

5. Deployment Diagram (Development Environment Model)

✓ Teacher Correction Applied: 3D Nodes, :Ethernet Hub, «deploy» arrows & component boxes.

deployment Development Environment Model

Development Environment Model



UML Deployment Architecture Legend & Mapping:

- :Ethernet Hub: Central network routing gateway connecting client and cloud server hardware nodes.
- 3D Nodes (DEV001 & DEV002): Physical/virtual hardware devices with CPU, RAM, and Disk controller specifications.
- «deploy» & Resides: Formal deployment relationships indicating software execution runtime and resident artifacts.

Fig 5: Deployment Diagram — Development Environment Model matching reference architecture.